

Course: Calculus I

ID-number: _____

Professor: Fco. Javier Vazquez Tavares

Date: 24/03/2026

Last Names and Names: _____

Group: 201

"In accordance with Tec de Monterrey Student Cod of Honor, in this exam I agree to conduct myself guided by academic honesty and not to use unauthorized or illegal means to do so."

Signature: _____

I Multiple option section. Read the following questions and identify the option that best answers each, then write the letter on the line.

1. (4 points) _____ Identify the quotient rule for derivatives.

(a) $\left(\frac{u}{v}\right)' = \frac{v'u - u'v}{u^2}$ (b) $\left(\frac{u}{v}\right)' = \frac{u'v - uv'}{v^2}$ (c) $\left(\frac{u}{v}\right)' = \frac{uv' - u'v}{v^2}$ (d) $\left(\frac{u}{v}\right)' = \frac{u'v - v'u}{u^2}$.

2. (4 points) _____ Identify the product rule for derivatives

(a) $(uv)' = u' + v'$ (b) $(uv)' = u'v + uv'$ (c) $(uv)' = uv' + uv'$ (d) $(uv)' = u'v + u'v$

3. (4 points) _____ Identify the chain rule for derivatives

(a) $[u(v(x))]' = v'(u(x))$ (b) $[u(v(x))]' = v'(u(x))u'(x)$ (c) $[u(v(x))]' = u'(v(x))v'(x)$ (d) $[u(v(x))]' = u'(v(x))$

4. (4 points) _____ Which of the following derivatives is correct

(a) $\frac{d}{dx}n^x = n^x \ln(n)$ (b) $\frac{d}{dx}n^x = \frac{1}{x}n^x$ (c) $\frac{d}{dx}n^x = xn^x$ (d) $\frac{d}{dx}n^x = ne^{x-1}$

5. (4 points) _____ Which of the following derivatives is correct

(a) $\frac{d}{dx}\sin(x) = \cos(x)$ (b) $\frac{d}{dx}\sin(x) = -\sin(x)$ (c) $\frac{d}{dx}\sin(x) = \sin(x)$ (d) $\frac{d}{dx}\sin(x) = -\cos(x)$

6. (4 points) _____ Which of the following derivatives is correct

(a) $\frac{d}{dx}\cos(x) = \cos(x)$ (b) $\frac{d}{dx}\cos(x) = -\sin(x)$ (c) $\frac{d}{dx}\cos(x) = \sin(x)$ (d) $\frac{d}{dx}\cos(x) = -\cos(x)$

7. (4 points) _____ Which of the following derivatives is correct

(a) $\frac{d}{dx}\log_n(x) = \frac{1}{\ln(x)}$ (b) $\frac{d}{dx}\log_n(x) = \frac{1}{x \ln(n)}$ (c) $\frac{d}{dx}\log_n(x) = \frac{\ln(x)}{x}$ (d) $\frac{d}{dx}\log_n(x) = \frac{1}{x \ln(x)}$

8. (4 points) _____ Which of the following options is the implicit derivative of $y^2 = x$

(a) $\frac{dx}{dy} = 0$ (b) $\frac{dx}{dy} = 2y - 1$ (c) $\frac{dx}{dy} = 1$ (d) $\frac{dx}{dy} = 1 - 2y$

II Implicit derivatives. Use the rules of derivatives that can be usefull to compute the implicit derivatives of the following functions. Compute $\frac{dy}{dx}$ and $\frac{dy}{dx}$.

1. (10 **points**) Compute the implicit derivative of the function $e^{x+y} = x^2y^3$.

2. (10 **points**) Compute the implicit derivative of the function $x^3 + y^3 = 6xy$.

III Critical points. In this section your task is to find the critical points of implicit derivatives considering only $\frac{dy}{dx} = 0$.

1. (13 points) Find the critical points of $x^{2/3} + y^{2/3} = 4^{2/3}$

2. (13 points) Find the critical points of $\frac{(x-3)^2}{9} - \frac{(y-8)^2}{4} = 1$

IV High order derivatives. Compute the second order derivative of the following functions and reduce them as much as possible.

3. (11 points) $f(x) = e^{x+3}$

4. (11 points) $h(x) = e^{3x} \sin\left(\frac{\pi}{3}x\right)$